16	**(a)**		Resolve down the plane $2g\sin 20° = 2a$ $a = g\sin 20° [= 3.352]$ $s = ut + \frac{1}{2}at^2$ with $s = 0.7,\ u = 1.4$ and their a $0.7 = 1.4t + \frac{1}{2}(g\sin 20°)t^2$ $(4.9\sin 20°)t^2 + 1.4t - 0.7 = 0$ Time taken is 0.352 s	**M1** **A1** **M1** **A1** **A1** **[5]**	**3.1b** **3.3** **3.3** **3.4** **3.2a**	N2L with no other forces with attempt to resolving the weight Allow $a = -g\sin 20°$ if it is clear that up the slope is the positive direction. Using *suvat* equation(s) leading to an equation for t; allow sign errors Correct equation using their $0.5a$ Allow coefficient of t^2 1.7 or better Cao Must select correct root if two roots given	Allow sin/cos interchange for M mark Solution BC is sufficient Allow if positive root only seen
16	**(b)**		Friction (and/or air resistance) would have the effect of slowing the particle so ignoring friction underestimates time	**B1** **[1]**	**3.5a**	Needs to indicate why ignoring resistance produces an underestimate	eg "friction will slow it down" for B1
16	**(c)**		$s = 0.7,\ u = 1.4,\ t = 0.8 \Rightarrow 0.7 = 1.4\times 0.8 + \frac{1}{2}a\times 0.8^2$ $a = \frac{0.7 - 1.12}{0.32} = -1.3125$ (–1.31 to 3 sf)	**M1** **A1** **[2]**	**3.3** **1.1b**	Use of *suvat* equation(s) to find a; allow sign errors Allow $a = 1.3125$ if sign convention clear	eg both u and s negative
16	**(d)**		N2L down the plane $2g\sin 20° - F = 2\times(-1.3125)$ $(F = 9.32859...)$ Resolve perpendicular to the plane $R = 2g\cos 20°$ $(R = 18.4\ldots)$ Use of $F = \mu R$ $\mu = \frac{9.32859...}{18.4...} = 0.506$ (3sf)	**M1** **A1** **M1** **A1** **M1** **A1** **[6]**	**3.1b** **1.1b** **3.3** **1.1b** **3.4** **1.1b**	All forces present and weight resolved; allow sin/cos interchange Fully correct equation; F need not be evaluated here No extra forces Fully correct equation; R need not be evaluated here Allow for their F and R used Accept 0.506 or 0.507 Must be 3sf	Notice R may be found first. Allow sin/cos interchange in *their* first equation but the second equation must be consistent to award M1